CS2 Week 7
Empty

Nonlinear Systems, Lyapunov Stability, Feedback Linearization

Exercise Classes

- Shared exercise with Haixiao and Kissan
- Theory recap
- Notes available on Kissan's website: https://kissanv.github.io

& on Haixiao's Polybox: https://polybox.ethz.ch/index.php/s/AReA9YBeJQKxBDn

- Quizzes, Examples & Old exam questions

Kissan's Website

Haixiao's Polybox

Course Schedule

Subject	Week
Intro – Discrete Time Systems	1
Diagonalization, Modal Coordinates, Contr/Obs.	2
State Feedback, Pole Placement, LQR	3
State Estimation, Luenberger Obs., Seperation Principle, LQG	4
DOFC with integral action, Intro to MIMO systems	5
Norms	6
Nonlinear Systems, Lyapunov, Feedback Linearization	7
break	8
Discrete-time Unconstrained Optimal Control	9
Convex Optimization	10
Model Predictive Control	11
Practical Model Predictive Control	12
Robust Model Predictive Control	13
Implementation Methods	14

Today

1. Nonlinear System
2. Lyapunov Stability
3. Feedback Linearization

1. Nonlinear Systems

Nonlinear Systems

- So far we've looked exclusively at linear systems
- An essential property of our linear system is superposition: $G(u_1 + u_2) = Gu_1 + Gu_2$
- Thanks to superposition, we could analyze the time response more easily and use the Laplace transform to study the system in the frequency domain

- Nonlinear systems do not satisfy superposition
- Their behavior depends on the input and initial condition
- Let's consider their state-space model:

linear system

$$\dot{x} = Ax(t) + Bu(t)$$
$$y = Cx(t) + Du(t)$$
$$x(0) = x_0$$

nonlinear system
time-invariant

$$\dot{x} = f(x(t), u(t))$$
$$y = g(x(t), u(t))$$
$$x(0) = x_0$$

nonlinear system

$$\dot{x} = f(x(t), u(t), t)$$
$$y = g(x(t), u(t), t)$$
$$x(0) = x_0$$

Nonlinear Systems

In general, nonlinear systems are more complex to analyze

Nonlinear systems can exhibit many interesting, though often counterintuitive, phenomena like:

or equilibrium subspace (if A singular)

- Multiple Equilibria: a linear system has 1 equilibrium (if A not singular)
  a nonlinear system can have 0, 1, or many equilibrium points

- Finite escape time: Unstable linear systems may diverge to infinity, but only in infinite time!
  Nonlinear systems may reach infinity in finite time

Problem 1 (Finite Escape Time)

Consider the nonlinear system:

$$\dot{x} = 1 + x^2.$$

Analyze the behavior of this system. Show that the system exhibits finite escape time and calculate the exact time t at which the state x goes to infinity, starting from an initial condition $x(t = 0) = x_0$.

assume $x_0 = 0$

$\dot{x} = 1 + x^2 = \frac{dx}{dt}$

$\int \frac{1}{1+x^2} dx = \int dt$

$t = \arctan(x) \rightarrow x(t) = \tan(t)$

$x \rightarrow \infty,\ t \rightarrow \pi/2$

Nonlinear Systems

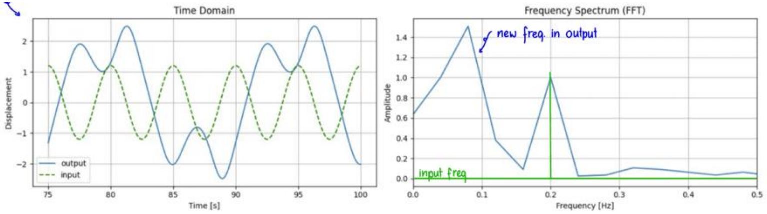
if there is an EW λ
↓ ith $Re(\lambda)=0$

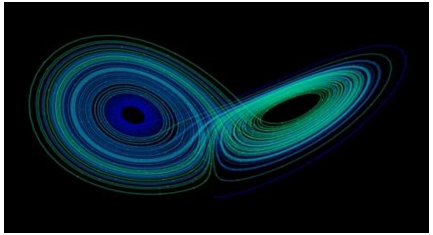
- Limit cycles: a linear system may generate permanent oscillations

  a nonlinear system may generate permanent oscillations independent from initial conditions

- Subharmonic Oscillations: a linear system will always output the same frequency as the input

  a nonlinear system can generate output frequencies that are submultiples or multiples of the input frequency





- Chaos: A nonlinear system can exhibit complex steady-state behavior that is neither an equilibrium nor a periodic oscillation

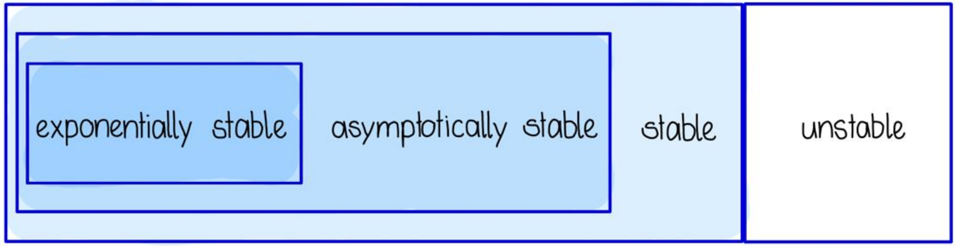
⇒ nonlinear system can exhibit any and all of the behaviors described!

Notions of Stability

- Define stability for a general autonomous system: $\dot{x} = f(x)$
- Stability is a property of an equilibrium point x_0, s.t. $\dot{x} = f(x_0) = 0$
- without loss of generality, we assume that $x_0 = 0$

bounded states

- An equilibrium point is …
  - **Lyapunov stable** if for each $\varepsilon \geq 0$, there is $\delta = \delta(\varepsilon) > 0$ s.t. $||x(0)|| < \delta \Rightarrow ||x(t)|| \leq \varepsilon$
  - **Asymptotically stable** if stable, and there exists $\delta > 0$ s.t. $||x(0)|| < \delta \Rightarrow lim_{t\to\infty}\, x(t) = 0$
    ↳ states converges to zero
  - **Exponentially stable** if there exist $\delta, \alpha, \beta > 0$ s.t. $||x(0)|| < \delta \Rightarrow ||x(t)|| < \beta e^{(-\alpha t)}$
    ↳ states decay expon. fast



Stability assessment is easier for linear systems (λ's of A), but no general test for nonlinear systems

2. Lyapunov Stability

▶ Lyapunov's stability theory is the **single most powerful method in stability analysis of non-linear systems**.

Lyapunov's Indirect Method

For arbitrary systems, there is no general way to determine stability.

- given a nonlinear system $\dot{x} = f(x)$ with eq. $x = 0$ $\rightarrow$ $A = \frac{\partial f}{\partial x}\Big|_{x=0}$
- the origin is asy. stable if, in its linearization $\dot{x} = Ax$, all eigenvalues of A are in the LHP
- this applies to the origin and a certain (possibly arbitrarily small) neighbourhood of the equilibrium point (if no eigenvalue with zero real part) $\rightarrow$ Hartman–Grobman theorem

⇒ Basically what we did in CS1 all along!

Lyapunov's Direct Method

could be phys. energy – doesn't have to
but will always mimic energy

- Idea: analyze stability of an equilibrium point x_0 with an "energy-like" function
- definitions & conditions: closed & bounded
  - $D \subset R^n$ is a compact set that contains the equilibrium point x_0
  - $V: D \rightarrow R$ is a smooth scalar function, called a Lyapunov candidate

in sense of Lyapunov

to be a valid LF

- The eq. point is stable if there exists a function $V(x)$, s.t. : Energy > 0 and at eq. lowest Energy

1. *positive definiteness:* $V(x) \geq 0, \forall x \in D$ and $V(x_0) = 0$
2. *non-increasing along trajectories:* $\dot{V}(x) = dV(x)/dx \cdot dx(t)/dt \leq 0, \forall x(t) \in D$

over time energy decreases

- stronger forms of stability
  - **asymptotic stability:** if $\dot{V} = 0$ only when $x(t) = x_0$
  - **exponential stability**: if $\dot{V} < -\alpha V$ for $\alpha > 0$

$\Rightarrow$ sufficient not necessary conditions for stability

just because we can't find Lyapunov function, doesn't mean system is unstable

Lyapunov Equation

- there is no general method of finding a Lyaponov function for arbitrary systems
- for a linearized sys. $\dot{x} = Ax$ one can always find a quadratic Lyapunov function

$V(x) = x^T P x$, P: symmetric, pos. definite $\rightsquigarrow$ fulfills $V > 0$

take deriv.

$\dot{V}(x) = \dot{x}^T P x + x^T P \dot{x}$ $\quad \dot{V}(x) = x^T A^T P x + x^T P A x = x^T (A^T P + PA) x = \dot{V}(x)$

$\longrightarrow$ for $\dot{V} \leq 0$, $A^T P + PA$ must be neg. def. $\rightarrow$ $\boxed{(A^T P + PA) = -\underbrace{Q}_{\text{pos. def.}}}$

$\Rightarrow$ Always has a solution if eigenvalues of A are in LHP!

Exam Question

$$\dot{V} = x^T Q x$$
$$A^T P + PA = -Q$$
$$V = x^T P x$$

Question 28 *Mark the correct answer. (2 Points)*
Consider the following LTI system

$$\dot{x} = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

The derivative of the Lyapunov function is given by:

$$\dot{V}(x) = -x_1^2(t) - 2x_2^2(t).$$

Find the corresponding Lyapunov function $V(x)$.

A $V(x) = x_1^2(t) + 2x_1(t)x_2(t) + 2x_2^2(t)$.

B $V(x) = 2x_1^2(t) + 2x_1(t)x_2(t) + x_2^2(t)$.

C $V(x) = 2.5x_1^2(t) + 2x_1(t)x_2(t) + x_2^2(t)$.

D $V(x) = x_1^2(t) + 2x_1(t)x_2(t) + x_2^2(t)$.

E $V(x) = 2.5x_1^2(t) + x_2^2(t)$.

F $V(x) = 2x_1^2(t) + 2x_2^2(t)$.

Question 28 *Mark the correct answer. (2 Points)*

Consider the following LTI system

$$\dot{x} = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

The derivative of the Lyapunov function is given by:

$$\dot{V}(x) = -x_1^2(t) - 2x_2^2(t).$$

Find the corresponding Lyapunov function $V(x)$.

$$\dot{V} = x^T Q x$$
$$A^T P + PA = -Q$$
$$V = x^T P x$$

LaSalles Invariance Principle

sometimes it is difficult to find $\dot{V} < 0$ <u>for asy. stability</u>; relax it to $\dot{V} \leq 0$ with LaSalle

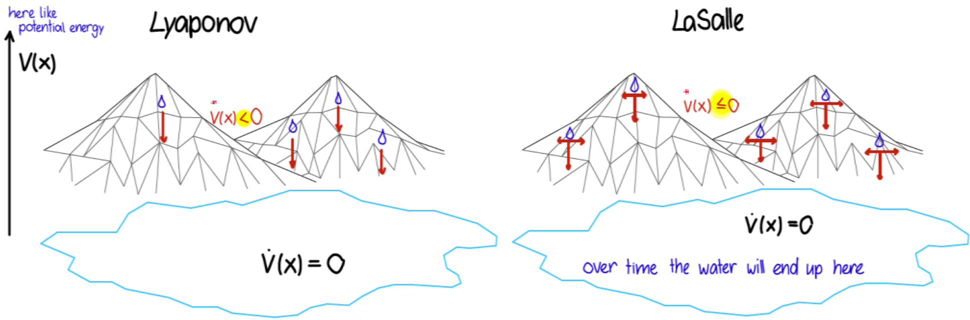
outside of x_0

Theorem (LaSalle's invariance principle)

Consider a system $\dot{x} = f(x)$. Let $\Omega \in D$ be a (compact) positively invariant set, i.e., a set such that if $x(t_0) \in \Omega$, then $x(t) \in \Omega$ for all $t \geq t_0$. Let $V : D \to \mathbb{R}^n$, such that $\dot{V}(x) \leq 0$ for all $x \in \Omega$. Then, $x(t)$ will eventually approach the largest positively invariant set in which $\dot{V} = 0$.

Intuitively:

Under **strict** Lyapunov conditions, the water always flows strictly downhill, so the energy $V(x)$ keeps decreasing until it reaches an equilibrium.
With **LaSalle**, the water may also move sideways on flat regions, so sometimes $\dot{V}(x) = 0$ even though it has not stopped yet.
Still, because it can never flow uphill, it will eventually end up in the **largest invariant flat region** where it can remain forever.



Example

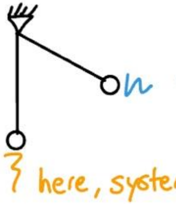
$$ml^2\ddot{\theta} = mgl\sin\theta - kd\,\dot{\theta}$$

- let's consider this pendulum
- intuitively we know, that it will go to $\theta = 0$, we now want to prove this asy. stable behaviour
- use Mech. energy as LF: $V(\theta,\dot{\theta}) = \frac{1}{2}ml^2\dot{\theta}^2 + mgl(1+\cos\theta)$
- taking d/dt → $\dot{V} = -\frac{kd}{ml^2}\dot{\theta}^2$, so $\dot{V}=0$ if $\dot{\theta}=0$ (indep. from θ)

→ $\dot{V}$ can be 0, even if we're not at eq. ⇝ Lyapunov theorem for asy. stability fails

- LaSalle states that every trajectory starting in Ω converges to the largest invariant set contained in $E = \{(\theta,\dot{\theta}) \mid \dot{V}(\theta,\dot{\theta}) = 0\}$

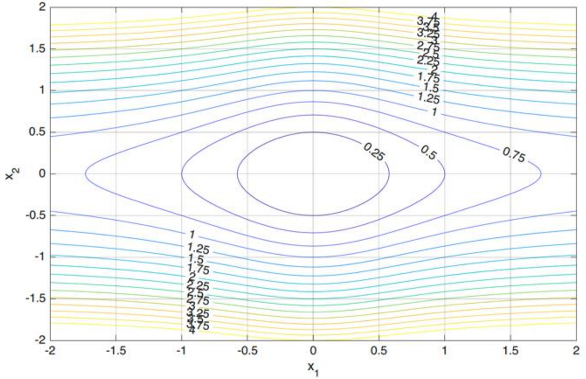
→ only trajectories where $\dot{V}$ remains $= 0$ are equilibria.



at the highest point $\dot{\theta}=0 \to \dot{V}=0$ but pend. will jump out of this state

here, system converges to it's true minimum.

Global Stability



- until now we only talked about local stability of an eq. point
- the extension to global stability is not difficult
- an eq. point is globally stable i.s.L if $D = R^n$, conditions (1) and (2) are fulfilled and if

$$\boxed{\textbf{(3):} \lim_{|x| \to \infty} V(x) = \infty}$$

⇝ the L. function needs to diverge

- then, we can say that the LF is radially unbounded
- the same extensions to asy. & expo. stability as before apply
- this result is called Barbashin-Krasovski-theorem

Control Lyapunov Functions

i.e. $\dot{x} = f(x) \rightarrow \dot{x} = f(x, u)$

- standard Lyapunov functions check stability of dynamical systems (without control input) $\longrightarrow$ extend it with control inputs

- Given $\dot{x} = f(x, u), f(0,0) = 0$, let $V(x)$ be positive definite and radially unbounded
- if there exists a $\tilde{u}(x)$ s.t. for every $x \neq 0$: $\dot{V}\big(x, \tilde{u}(x)\big) = \dfrac{\partial V}{\partial x} f(x, \tilde{u}) < 0$

then $V(x)$ is said to be a CLF

- this is essentially the same as before, except that we now have some control over $\dot{V}$ by choosing an appropriate $\tilde{u}$

$\rightarrow$ it could be that $\dot{V}(x, 0) \geq 0$ but $\dot{V}\big(x, \tilde{u}(x)\big) < 0$:

- ► Clearly, if a function is shown to be a CLF, it provides a way to construct a stabilizing feedback.
- ► **Artstein's theorem**: A dynamical control system has a differentiable control-Lyapunov function if and only if there exists a regular stabilizing feedback $u(x)$.

$\dot{V} < 0 \Leftrightarrow$ there exists stabil. $u(x)$

Exam Question

Question 26 *Mark all correct statements. (2 Points)*
Which of the following statements about nonlinear systems are true?

a Linear systems can go to infinity in finite time.

b Nonlinear systems can have multiple or no equilibrium points.

c In nonlinear systems, limit cycles are outputs at frequencies that are submultiples or multiples of the input frequency.

d In nonlinear systems the effect of inputs can be separated from the effect of initial condition.

B)

Exam Question

Valid LF →

- The eq. point is stable if there exists a function $V(x)$, s.t. :
  1. *positive definiteness:* $V(x) \geq 0$ and $V(x_0) = 0$
  2. *non-increasing along trajectories:* $\dot{V}(x) = dV(x)/dx \cdot dx(t)/dt \leq 0$

Question 27 *Mark all correct statements. (2 Points)*
Consider the following nonlinear system

$$\dot{x}_1(t) = -x_1(t) + 4x_2(t),$$
$$\dot{x}_2(t) = -x_1(t) - x_2^3(t).$$

The only equilibrium point of the system is at the origin ($x_1(t) = x_2(t) = 0$). Evaluate the stability of the origin using the Lyapunov function

$$V(x) = x_1^2(t) + 4x_2^2(t).$$

Which of the following statements are true?

a The origin is not Lyapunov stable.

b $V(x)$ is not a valid Lyapunov function.

c The origin is Lyapunov stable but not asymptotically stable.

d $V(x)$ is radially unbounded.

e The origin is a globally asymptotically stable equilibrium point.

Question 27 *Mark all correct statements. (2 Points)*
Consider the following nonlinear system

$$\dot{x}_1(t) = -x_1(t) + 4x_2(t),$$
$$\dot{x}_2(t) = -x_1(t) - x_2^3(t).$$

Valid LF →

- The eq. point is stable if there exists a function $V(x)$, s.t. :
1. *positive definiteness:* $V(x) \geq 0$ and $V(x_0) = 0$
2. *non-increasing along trajectories:* $\dot{V}(x) = dV(x)/dx \cdot dx(t)/dt \leq 0$

The only equilibrium point of the system is at the origin ($x_1(t) = x_2(t) = 0$). Evaluate the stability of the origin using the Lyapunov function

$$V(x) = x_1^2(t) + 4x_2^2(t).$$

Which of the following statements are true?

a The origin is not Lyapunov stable.

b $V(x)$ is not a valid Lyapunov function.

c The origin is Lyapunov stable but not asymptotically stable.

d $V(x)$ is radially unbounded.

e The origin is a globally asymptotically stable equilibrium point.

Exam Question

Question 29 *Mark all correct statements. (2 Points)*
Which of the following statements about Lyapunov functions are true?

A) LaSalle's invariance principle is useful in cases where there are points other than the equilibrium where the Lyapunov function is zero, i.e., $V(x) = 0$. $\dot{V} = 0!$

B) A common candidate for a Lyapunov function of a nonlinear system are quadratic Lyapunov functions derived from linearization of the system near equilibrium points and solving the Lyapunov equation.

$$\boxed{\begin{gathered}\dot{V} = x^T Q x \\ A^T P + PA = -Q \\ V = x^T P x\end{gathered}}$$

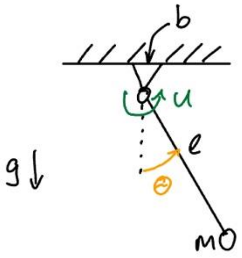
3. Feedback Linearization

Intuition

- Previously we learned how to analyze stability of nonlinear systems
- Now we're looking at how we can implement a feedback loop for a nonlinear system in a clever way
- this involves transforming the nonlinear system into one that behaves linear with a **coordinate transformation**

⇒ The technique we apply to get this effect is called: <u>Feedback Linearization</u>

Example



- Input: Torque u
- Output: State θ
- Dynamics (normalized):

$$\ddot{\theta} + c\dot{\theta} + a\sin(\theta) = u$$

$a = g/l$; $c = b/I$; $u = \tau/I$

Write as 1st order sys. $\longrightarrow$ $x_1 = \theta$, $x_2 = \dot{\theta}$

$$\boxed{\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -a\sin(x_1) - cx_2 + u \\ y &= x_1 \end{aligned}}$$

clearly a nonlinear system!

- The trick now is how we choose the input $\boldsymbol{u}$ $\Rightarrow$ introduce a virtual control input $\boldsymbol{v}$

choose

$v =$

substitute $\longrightarrow$

$$\boxed{\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= v \\ y &= x_1 \end{aligned}}$$

coordinate change cancels out nonlinearities

- The actual physical input remains $\boldsymbol{u}$, we write it as:

$$u = v + a\sin(x_1) + c\,x_2$$

↳ v is virtual we can choose how we like...

Example

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = v$$
$$y = x_1$$

$u = v + a\sin(x_1) + c\,x_2$

↳ v is virtual we can choose how we like…

- Design a stabilizing feedback law: $v = -Kx = -k_1 x_1 - k_2 x_2$

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -k_1 x_1 - k_2 x_2$$
$$y = x_1$$

$\xrightarrow{\text{write in matrix form}}$ $A_{cl} = \begin{bmatrix} 0 & 1 \\ -k_1 & -k_2 \end{bmatrix}$

⇒ Eigenvalues can be chosen freely!

Can't we just use this for any nonlinear system?

- Designing a control law like this requires some assumptions:

1. We know the system well enough to allow for state feedback
2. The system must have a certain structure

→ Systems to which we can apply feedback linearization have a geometric property called…

Differential Flatness

⇒ iff a system is differential flat, we can apply feedback linearization

Def.: A system $\dot{x} = f(x, u)$
$y = h(x, u)$ ⇒ "flat output"

is differentially flat, if one can compute all x and u as a function of y and derivatives of y

⇒ there is a mapping, s.t. $(x, u) \equiv \left(y, \dot{y}, \dots, y^{(m)}\right), m \mathrel{\widehat{=}}$ finite

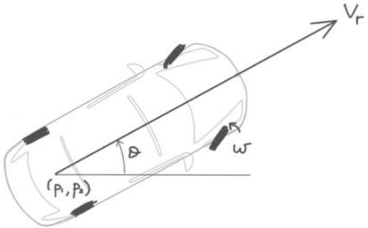
Note: the flat output y is not necessarily the system output

Example: is the pendulum differentially flat?

$$\boxed{\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -a\sin(x_1) - cx_2 + u \\ y &= x_1 \end{aligned}} \xrightarrow{\text{reconstruct } x, u \text{ using comb. of } y, \dot{y}, \ddot{y}, \dots} \boxed{\begin{aligned} x_1 &= y \\ x_2 &= \dot{y} \\ u &= \ddot{y} + c\dot{y} + a\sin(y) \end{aligned}}$$

works! system is differentially flat

Last Example



- Dynamics:

$$\dot{p}_1(t) = v_r(t)\cos\theta(t),$$
$$\dot{p}_2(t) = v_r(t)\sin\theta(t),$$
$$\dot{\theta}(t) = \frac{v_r(t)}{b}\tan w(t)$$

speed of rear axle

orientation

states: p_1, p_2, θ inputs: v_r, w ← steering angle

coordinates

- Check whether all states and inputs can be expressed through a flat output and its derivatives

→ If yes, the system is differentially flat

⇒ it turns out the system is flat with the flat output (p_1, p_2):

we can write:

$$\theta = \arctan\left(\dot{p}_2/\dot{p}_1\right) \xrightarrow{d/dt}$$

$$v_r = \sqrt{\dot{p}_1^2 + \dot{p}_2^2}$$

$$w = \arctan\frac{b\dot{\theta}}{v_r} = \arctan\left(\frac{b}{\sqrt{\dot{p}_1^2 + \dot{p}_2^2}} \cdot \frac{\ddot{p}_2\dot{p}_1 - \dot{p}_2\ddot{p}_1}{\dot{p}_1^2 + \dot{p}_2^2}\right)$$

The fact that this construction is possible shows diff. flatness

⇒ By knowing only the car's position over time, we can reconstruct all states and inputs at all times

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.